Smart ATM Security System using AI (YOLO)

1. Introduction

Automated Teller Machines (ATMs) are widely used for banking transactions such as cash withdrawal, deposits, and balance inquiries. However, ATMs are frequently targeted by criminals for robbery, vandalism, and physical damage. Traditional security systems like CCTV cameras only record incidents but cannot automatically detect or prevent suspicious activities in real time.

To address this limitation, this project proposes a **Smart ATM Security Monitoring System using Artificial Intelligence and Computer Vision**. The system uses a trained **YOLO (You Only Look Once) object detection model** to analyze live video feeds and detect suspicious activities such as fights, weapons, fire, or smoke inside or near the ATM booth.

When such activities are detected, the system automatically classifies the threat level and sends alerts to the monitoring authority through a web dashboard. This approach enables **real-time monitoring, automated threat detection, and faster response**, improving the security of ATM installations.

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2. Objective of the Project

The main objective of this project is to develop an **AI-based intelligent monitoring system for ATM security** that can automatically detect suspicious activities using computer vision.

Specific objectives include:

- To monitor ATM surroundings using real-time video feed.
- To detect suspicious activities such as **fight, weapon usage, fire, or smoke** using a trained YOLO model.
- To classify threats into **Normal, Minor Threat, and Major Threat** categories.
- To provide **real-time alerts and monitoring through a web dashboard**.
- To store incident logs for security analysis and investigation.
- To reduce dependency on manual monitoring and improve response time.

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3. Problem Statement

Most ATM security systems rely on CCTV cameras that only record footage. These systems require human operators to constantly monitor the cameras, which is inefficient and prone to delays. In many cases, suspicious activities such as ATM robbery or vandalism are detected only after the incident has already occurred.

Additionally, traditional systems cannot automatically identify dangerous situations such as fire outbreaks, weapon presence, or violent activity.

Therefore, there is a need for an **intelligent automated surveillance system** that can analyze video feeds in real time and instantly detect abnormal or dangerous situations. This project addresses this issue by integrating **Artificial Intelligence, Computer Vision, and Machine Learning** to create a smart ATM security monitoring solution.

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4. Technologies Used

The system is implemented entirely using software technologies.

Artificial Intelligence and Machine Learning

- **YOLO (You Only Look Once)** object detection model
- Real-time object detection and classification

Computer Vision

- **OpenCV** for image processing and video frame handling

Programming Language

- **Python**

Backend Framework

- **Flask / FastAPI** for server-side processing

Frontend Development

- **React.js / HTML / CSS / Tailwind**

Database

- **MongoDB** for storing logs and alerts

Libraries Used

- Ultralytics YOLO
- OpenCV
- NumPy
- Pandas

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5. YOLO Model in the Project

YOLO (You Only Look Once) is a deep learning-based object detection algorithm that can detect multiple objects in an image in real time. Unlike traditional detection methods, YOLO processes the entire image in a single pass, making it extremely fast and suitable for real-time surveillance systems.

In this project, the YOLO model is trained on a dataset containing images of:

- Fighting activities
- Weapons such as guns or knives
- Fire or flames
- Smoke
- Normal human presence

The trained YOLO model analyzes each frame from the video feed and identifies objects with bounding boxes and confidence scores.

Example detection output:

weapon detected – confidence 0.91
fire detected – confidence 0.88
person detected – confidence 0.96

Based on these detections, the system determines the level of threat.

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6. System Architecture

The system architecture consists of several modules that work together to monitor ATM security.

Video Input Layer

The system receives real-time video input from a webcam or CCTV feed that monitors the ATM booth.

Detection Layer

Each video frame is processed using the YOLO model to detect suspicious objects or activities.

Threat Analysis Layer

Detected objects are analyzed to determine the severity of the situation.

Threat classification:

- **Normal Condition** – Normal ATM usage
- **Minor Threat** – Suspicious movement or fight detected
- **Major Threat** – Weapon, fire, or smoke detected

Alert and Response Layer

When a threat is detected, the system generates alerts and sends them to the monitoring dashboard.

Monitoring Dashboard

A web-based dashboard displays:

- Live camera feed
- Detected threats
- Alert notifications
- Event history logs

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7. Working Methodology

The working of the system follows several steps:

Step 1 – Video Capture

The system captures real-time video using a webcam or CCTV feed.

Step 2 – Frame Processing

The video stream is divided into frames, and each frame is processed using OpenCV.

Step 3 – Object Detection using YOLO

The YOLO model analyzes each frame to detect objects such as:

- Person
- Weapon
- Fire
- Smoke
- Fight

Step 4 – Threat Classification

Based on detected objects, the system determines the threat level.

Example logic:

- If **weapon detected → Major Threat**
- If **fire or smoke detected → Major Threat**
- If **fight detected → Minor Threat**
- If only normal person detected → Normal Condition

Step 5 – Alert Generation

When a threat is detected, the system generates alerts and stores the event in the database.

Step 6 – Dashboard Visualization

The dashboard displays alerts, system status, and historical logs for administrators.

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8. Features of the System

The proposed system provides several advanced features:

- Real-time video surveillance
- AI-based threat detection
- Automatic classification of suspicious activities
- Web-based monitoring dashboard
- Real-time alert notifications
- Event logging and analytics
- Scalable and cost-effective implementation

9. Expected Outcomes

The system is expected to provide the following benefits:

- Early detection of suspicious activities near ATM machines
- Faster response to security threats
- Reduced reliance on manual monitoring
- Improved safety of ATM installations
- Automated security monitoring system using AI

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10. Conclusion

The Smart ATM Security System using AI and YOLO demonstrates how modern technologies such as **Artificial Intelligence, Computer Vision, and Machine Learning** can be applied to improve ATM security. By automatically analyzing real-time video feeds and detecting suspicious activities, the system provides a proactive approach to ATM monitoring.

The integration of YOLO-based object detection with a real-time monitoring dashboard allows security personnel to respond quickly to potential threats. This project highlights the potential of AI-driven surveillance systems in enhancing security infrastructure and preventing criminal activities.